Finite Element Analysis

Prosthetic Running Blade

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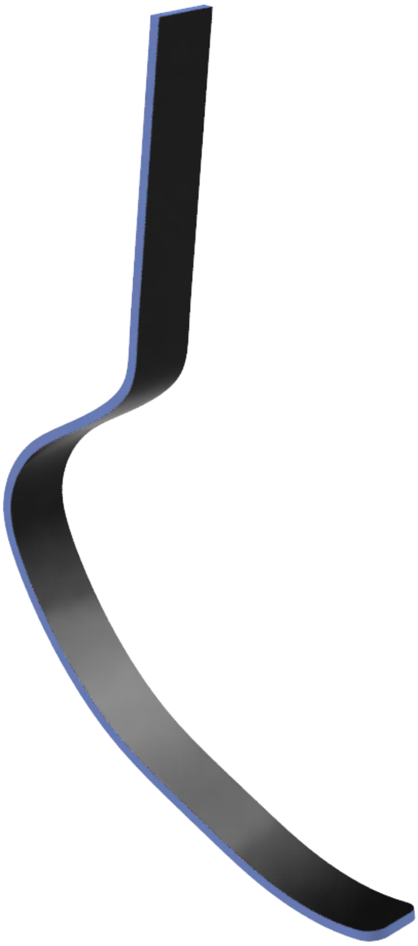

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Contents

1 Introduction

1.1 Project Objective

Prosthetic running blades let leg amputees compete in sprinting and jumping events. These blades work as springs, returning energy on push-off. The brief [1] required an initial blade to be designed and analysed in carbon-fibre reinforced composite (CFRP) against three criteria:

- Should be lightweight, defined as below 1kg.
- Should be able to withstand a minimum of 1,000,000 loading cycles.
- Should have a fundamental frequency no lower than 50Hz.

1.2 Blade Design

The geometry was inspired by the Össur Cheetah Xtreme [2]. A posterior sprinting blade designed for 100-200m, characterised by a tight curve and a longer flat toe.

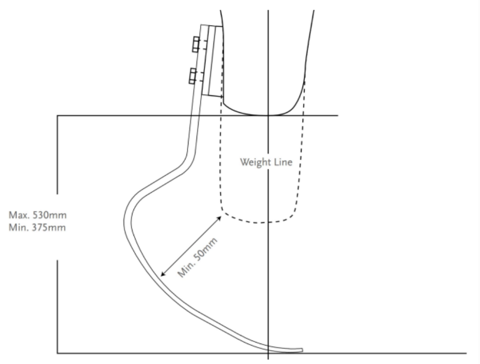

Figure 1: Össur Cheetah Xtreme Diagram [2]

The simplified CAD was created using the sweep tool with a centreline and profile sketch, resulting in a curved blade with a uniform thickness of 7mm, a width of 60mm, and a total height of 530.7mm. The toe was rounded with a 12.5mm fillet radius.

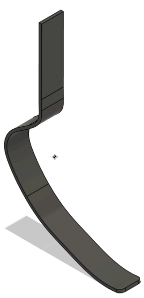

Figure 2: Initial Design CAD.

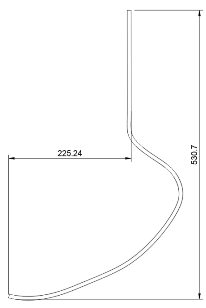

Figure 3: CAD Dimensions

2 Methods

2.1 Boundary Conditions and Loads

The blade would be fixed where the socket attaches; this was modelled as a fixed support on the interior face of the vertical section.

Two forces were applied to the underside of the toe [1]. A normal force of 2100 N was applied in the Y-direction, representing approximately 3× the bodyweight of a 70 kg person at peak foot-strike, as is typical of reaction force during running. A friction force of 500 N was applied in the X-direction. This represents the foot-strike phase of the running cycle, where the blade first contacts the track and friction acts backwards, resisting the forward motion of the foot.

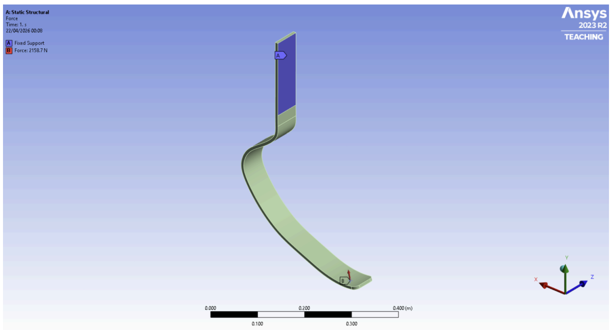

Figure 4: Fixed Support and Force Conditions

2.2 Materials

CFRP is anisotropic, but modelling this properly requires additional tools that weren't used in this analysis. Instead, the blade was modelled as if CFRP were isotropic. Properties were taken from a published study on prosthetic blades using the same composite [3]. This is a simplification, and the stress results should be read with that in mind.

Property	**Value**	**Units**
Density	1.6	g/cm^{-3}
Young's Modulus	70000	MPa
Poisson Ratio	0.1	–
Tensile Yield Strength	600	MPa
Compressive Yield Strength	570	MPa
Tensile Ultimate Strength	600	MPa
Compressive Ultimate Strength	570	MPa

Table 1: Material properties of CFRP

Aluminium 2014-T4 was specified in the brief but was not available in Ansys 2023's library. The built-in Aluminium Alloy was used instead [4]. This would introduce some inaccuracy, as the properties are not identical; the aluminium results should therefore be treated as approximate.

Property	Value	Units
Density	2.77	g/cm^{-3}
Young's Modulus	71000	MPa
Poisson Ratio	0.33	–
Tensile Yield Strength	280	MPa
Compressive Yield Strength	280	MPa
Tensile Ultimate Strength	280	MPa

Table 2: Material properties of Aluminium Alloy

2.3 Fatigue Analysis Method

For CFRP, no S-N curve was used as no universally applicable curve exists. Instead, a safety factor of 3 was applied to the maximum principal stress, as this is the appropriate failure criterion for brittle materials. If the result is below the CFRP tensile ultimate strength, the design passes the fatigue criteria.

For aluminium, a full fatigue analysis was performed using the built-in S-N curve, which defines the relationship between stress and the number of cycles to failure. The fatigue tool was set up with the following:

Parameter	Value	Reasoning
Loading Type	Zero-Based	During a foot strike, the load goes from zero to max then back to zero. The direction does not reverse, making zero based (R=0) loading appropriate. [5]
Mean Stress Theory	Gerber	Gerber is the most appropriate for ductile materials such as aluminium [5]
Stress Component	Von Mises	Von Mises stress combines stresses in all directions into a single value, making it suitable for ductile materials such as aluminium [5]

Table 3: Fatigue Tool Settings

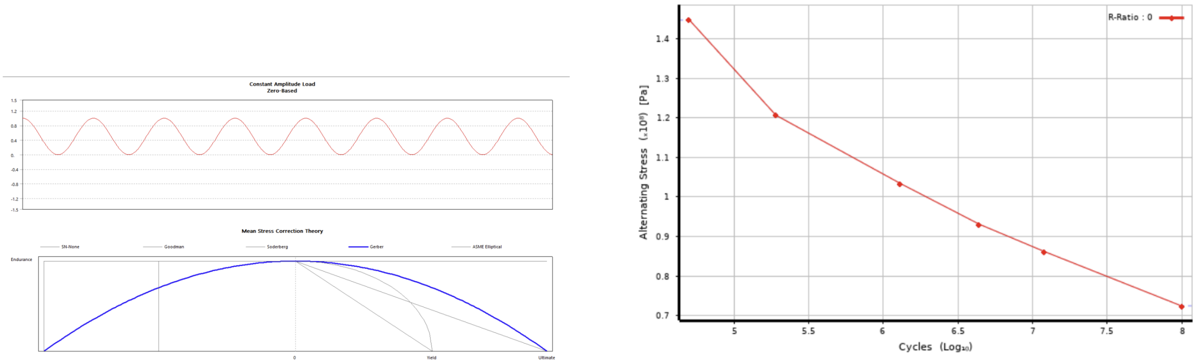

Figure 5: Fatigue Methods Used and S-N Curve used for Aluminium Alloy

2.4 Frequency Analysis Method

A modal analysis was done to find the fundamental frequency of each design. A direct solver was used, providing better accuracy and more stable convergence than iterative methods. The fundamental frequency must remain above 50 Hz. If the excitation frequency from foot-strike matches the blade's fundamental frequency, resonance occurs, and displacement grows until failure.

3 Results

3.1 Initial Design - CFRP

The initial design had a mass of 518.9 g, below the 1 kg lightweight criteria.

3.1.1 Mesh Refinement

The model needs to be broken down into a mesh of elements to be solved numerically. This introduces a discretisation error. Coarser meshes require less computational power but produce greater discretisation error; finer meshes reduce this error at a higher cost. A convergence study was carried out to determine the coarsest mesh at which the maximum stress and maximum deformation stop changing significantly (defined as 1%), balancing accuracy with solve time.

Two mesh quality metrics were tracked at each step: aspect ratio (element elongation) and Jacobian ratio (element distortion at curved or sharp features). Both have an ideal value of 1, and values further from 1 reduce solution accuracy.

Global Element Size (mm)	Nodes	Elements	Max Stress (MPa)	%Δ Stress	Max deformation (mm)	%Δ Deformation
20	1071	405	1033.5	N/A	377.6	N/A
10	2854	1178	978.9	-5.28%	397	+5.14%
5	9562	4425	991.9	+1.33%	398.5	+0.38%
4	15574	7625	994.5	+0.26%	398.5	0%
3	33405	18012	992.2	-0.23%	398.5	0%
2	70411	38076	992.1	-0.01%	398.5	0%

Table 4: Global Mesh Results

Global Element Size (mm)	Max AR	Average AR	Jacobian Ratio	Min Jacobian Ratio
20	20.93	5.17	0.925	0.43
10	32.1	2.49	0.968	0.7
5	15.129	2.3032	0.98	0.81759
4	19.002	2.2363	0.99146	0.82214
3	13.872	2.2751	0.99429	0.77603
2	9.4244	2.1032	0.99651	0.87658

Table 5: Global Mesh Quality Metrics

The maximum stress stabilises at a 5mm element size, with all later changes below 1%. Maximum deformation converges earlier at 10mm. Mesh quality improves with refinement; average aspect ratio decreases from 5.17 at 20mm to 2.2 at finer sizes, and average Jacobian ratio rises above 0.98; elements are becoming closer to their ideal shape. The maximum aspect ratio remains high everywhere; this is likely due to the thin shape forcing stretched elements, but it seems to be a local effect, as it is not reflected in the average. A global element size of 5mm was selected as the mesh, as it achieved convergence with good overall mesh quality at a reasonable cost.

The static analysis of the 5mm global mesh showed that stresses are concentrated on the exterior face of the main curve, where bending stress is highest. Local refinement was applied to this region to better solve the stress, without refining the entire model.

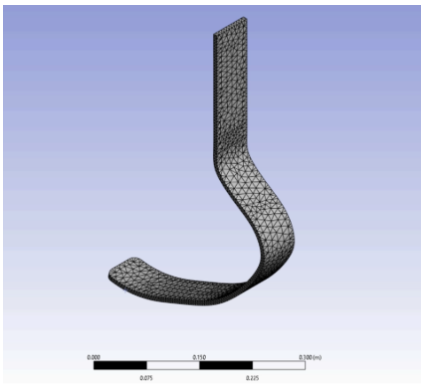

Figure 6: 5mm Mesh

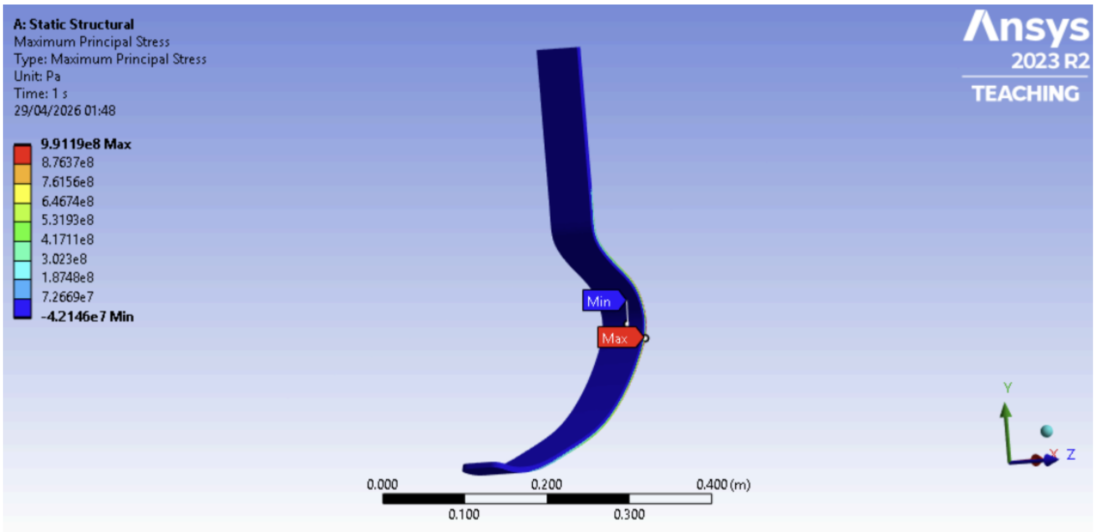

Figure 7: Maximum Principle Stress Plot - Front

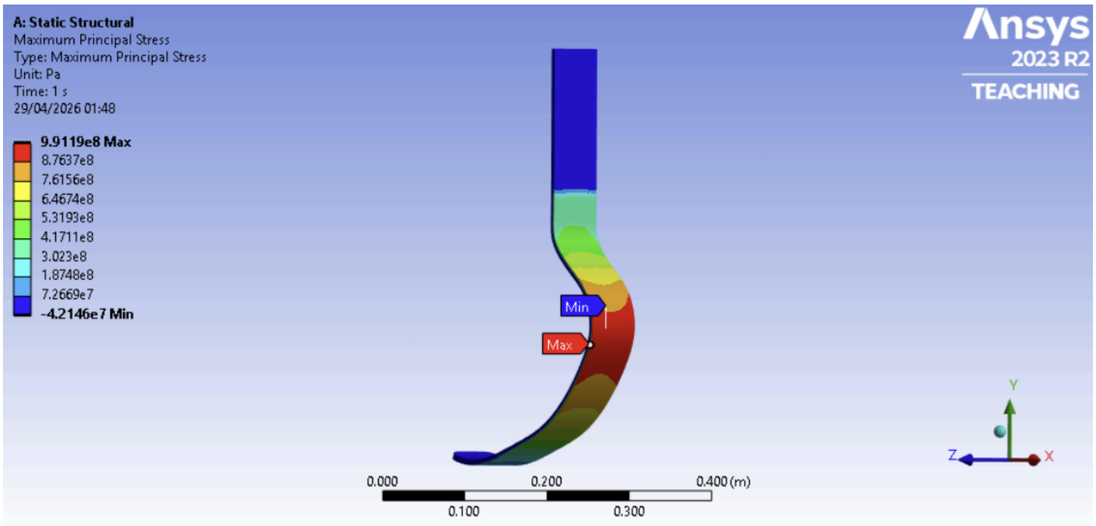

Figure 8: Maximum Principle Stress Plot - Back

The face was split in the geometry, and another study was conducted.

Global + Local Element Size (mm)	Total Nodes	Total Elements	Max Stress (MPa)	%Δ Stress	Max deformation (mm)	%Δ Deformation
5mm no local	9562	4425	991.9	N/A	398.5	N/A
5mm + 2mm	40053	21733	974.7	-1.73%	398.5	0%
5mm + 1mm	140524	80742	973.3	-0.14%	398.5	0%

Table 6: Global and local mesh results

Local refinement decreased the maximum stress by 1.73% and had no effect on deformation. Further refinement to 1 mm changed the stress by only -0.14%. The solution has settled, and global convergence was not masking a significant localised stress feature. The 5 mm + 2 mm mesh was selected as the final mesh for all subsequent analysis, as it captures the stress concentrations more accurately than the global mesh alone at a reasonable computational cost.

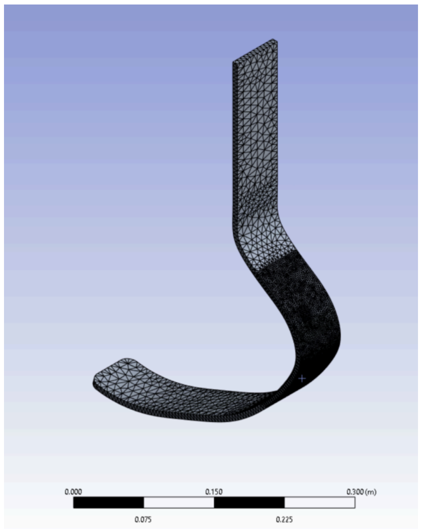

Figure 9: 5mm + 2mm Mesh

3.1.2 Deformation and Static Stress

The maximum deformation was 398mm, located at the toe. This deformation on a 530mm blade is unrealistic, as in practice the blade would have failed long before reaching this. This is a limitation of linear static FEA: the model assumes linear elastic behaviour and small deformations, and therefore reports displacement regardless of whether the material has exceeded its strength.

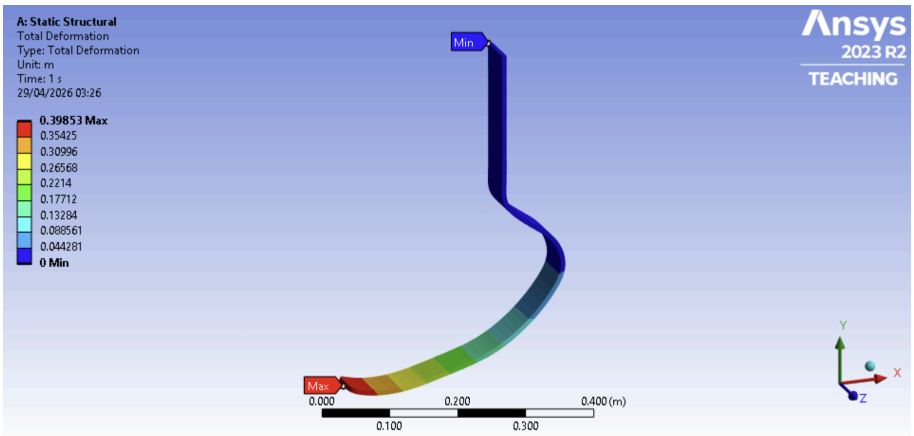

Figure 10: Deformation Plot

The maximum stress was 974.7 MPa, on the convex face of the curve, which is in tension under the applied loading. The concave face is in compression at -40.5 MPa. The mounting section and toe are also low-stress. This is consistent with classical beam theory: stress concentrates at the locations of maximum bending moment and on the surfaces furthest from the neutral axis.

A safety factor of 3 is applied to the maximum stress to assess fatigue.

$$974.7 \times 3 = 2924.1 \text{ MPa}$$

The effective design stress of 2924.1 MPa greatly exceeds the CFRP ultimate tensile strength of 600 MPa, indicating critical failure of the fatigue criteria.

3.1.3 Fundamental Frequency

A modal analysis was run on the 5 mm + 2 mm mesh with fixed support as the only boundary condition. No loads were applied since the analysis is independent of external forcing. Only the fundamental frequency was reported, as higher modes sit above the excitation range from foot-strike and aren't likely to be relevant.

The fundamental frequency (Mode 1) was 26.661 Hz, below the 50 Hz requirement. The initial design, therefore, also fails the frequency criteria. The Mode 1 shape is a flapping motion of the blade with the toe oscillating vertically.

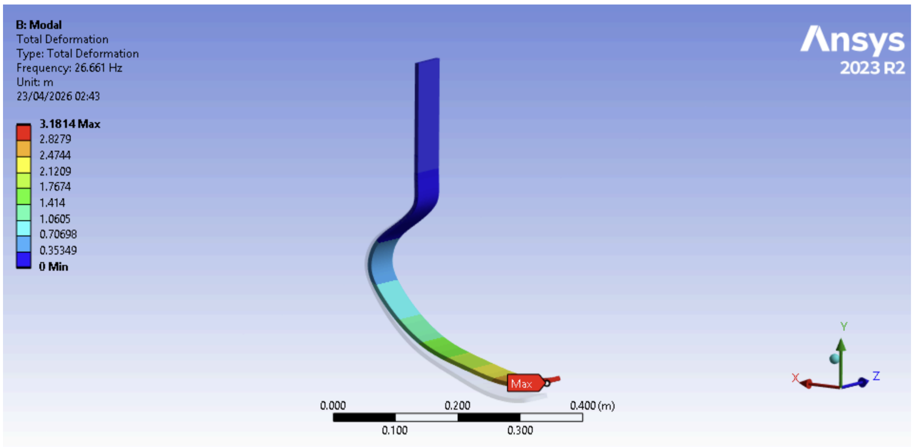

Figure 11: Modal Analysis - Mode 1

3.2 Iterated Design - CFRP

3.2.1 Design Change

The initial design failed both the fatigue and frequency criteria by a wide margin. The geometry was modified to increase the blade's bending stiffness, thereby reducing peak stress and increasing the fundamental frequency.

Three changes were made:

- Thickness increased from 7 mm to 12 mm.
- The width increased from 60 mm to 70 mm.
- Overall length reduced by 40 mm.

The new mass is 964.16 g, which just meets the lightweight criterion of under 1 kg.

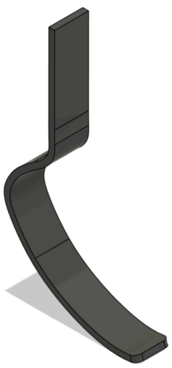

Figure 12: Iterated Design CAD

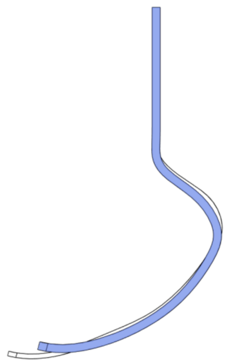

Figure 13: New vs Old CAD Diagram

The reasoning for the changes comes from beam bending theory. The second moment of area of a rectangular cross-section is:

$$I = \frac{bh^3}{12}$$

Symbol	Parameter
b	Width of cross section
h	Thickness in bending direction
I	Second moment of area

Table 7: Symbol definitions

Bending stress is inversely proportional to I, so increasing I reduces maximum stress. Going from h = 7 to 12 mm increases I by a factor of 5.038, and the width going from 60 to 70 mm adds a further 17%. The 40 mm length reduction primarily occurs along the x-axis, reducing the moment arm of the 2100 N load by about 7.5%. Since this is the dominant load, maximum bending stress drops by roughly the same amount.

The fundamental frequency of a cantilever tip mass is:

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3EI}{mL^3}}$$

Symbol	Parameter
fn	Natural or fundamental frequency
E	Youngs modulus
I	Second moment of area
m	Mass
L	Length along the beam

Table 8: Symbol definitions

This formula assumes a straight uniform cantilever with a concentrated mass at the tip. The blade is curved with distributed mass, so the calculation would slightly underestimate the frequency.

Taking the ratio of the redesign to the initial design.

$$\frac{I_2}{I_1} = 5.88 \qquad \frac{L_2}{L_1} = 1.27 \qquad \frac{m_2}{m_1} = 1.86$$

This predicts a frequency increase of approximately 2×, indicating the fundamental frequency will likely meet the 50 Hz criterion.

3.2.2 Mesh Refinement

Another convergence study was run.

Global Element Size (mm)	Nodes	Elements	Max Stress (MPa)	%Δ Stress	Max deformation (mm)	%Δ Deformation
20	1121	431	141.157	N/A	20.065	N/A
10	3424	1496	143.31	+1.53%	20.208	+0.71%
5	13895	7325	143.66	+0.24%	20.224	+0.08%
4	21043	11120	142.52	-0.79%	20.236	+0.06%
3	36597	19645	142.75	+0.16%	20.236	0%
2	91283	51610	142.17	-0.41%	20.236	0%

Table 9: Global Mesh Results

Global Element Size (mm)	Max AR	Average AR	Jacobian Ratio	Min Jacobian Ratio
20	14.835	3.2852	0.94414	0.54139
10	20.619	2.3152	0.97865	0.72068
5	10.664	2.2842	0.991	0.68666
4	9.1397	2.1303	0.99322	0.80742
3	8.9666	2.1819	0.99536	0.83091
2	14.971	2.2381	0.99711	0.85756

Table 10: Global Mesh Quality Metrics

Stress converges at a 5 mm global element size, and deformation at 10 mm. Mesh quality follows the same pattern as before, but is a bit better overall. The average aspect ratio drops to around 2.2 at finer sizes, and the Jacobian averages remain around 0.96 across all steps, with minimums above 0.68 from 5 mm onwards. The improvement is likely due to the thicker cross-section, which gives the mesh more room to fit cleaner elements through the geometry.

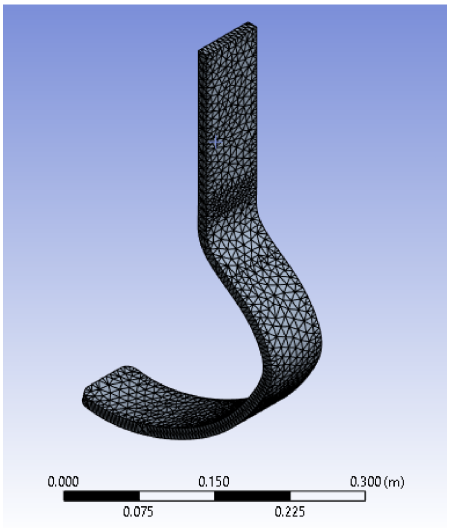

Figure 14: 5mm Mesh

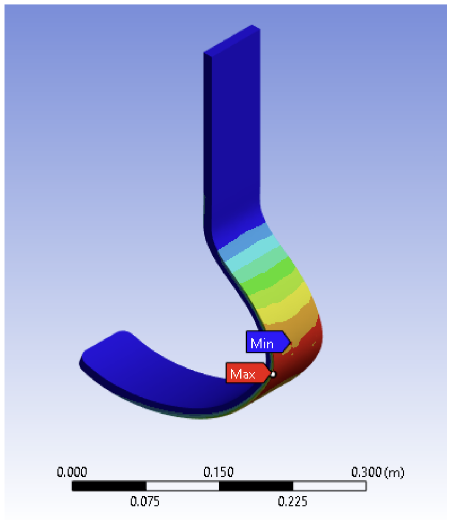

Figure 15: Max Principle Stress Plot

Local refinement was applied to the high-stress convex side of the curve, as in the initial design.

Global + Local Element Size (mm)	Total Nodes	Total Elements	Max Stress (MPa)	%Δ Stress	Max deformation (mm)	%Δ Deformation
5mm no local	13895	7325	143.66	N/A	20.224	N/A
5mm + 2mm	51496	29450	139.64	-2.80%	20.241	+0.08%
5mm + 1mm	174992	103559	139.3	-0.24%	20.241	0%

Table 11: Global and local mesh results

Local meshing to 2 mm dropped the maximum stress by 2.80%, with no significant change in deformation. This is a bigger step than in the initial design and above the 1% threshold, indicating the 5 mm global wasn't fully resolving the stress on the curve face. The new geometry seems to concentrate stress more sharply on the curve, so local meshing matters more here than before. The 5 mm + 2 mm local mesh was chosen for all further analysis.

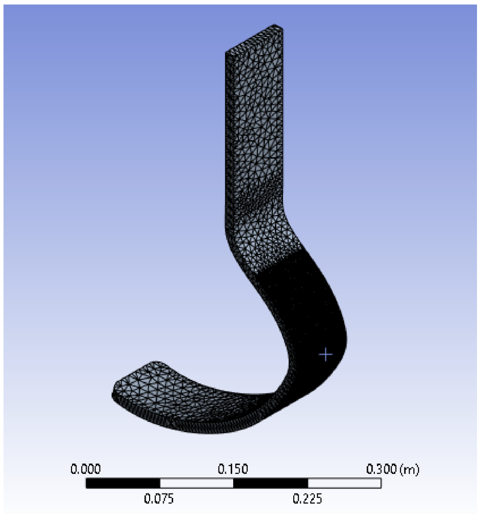

Figure 16: 5mm + 2mm Mesh

3.2.3 Deformation and Static Stress

The maximum deformation was 20.2 mm at the toe tip. This is within the small-deflection range where linear static FEA is valid; the result is realistic, unlike 398mm, which was far outside the limit and not physically meaningful.

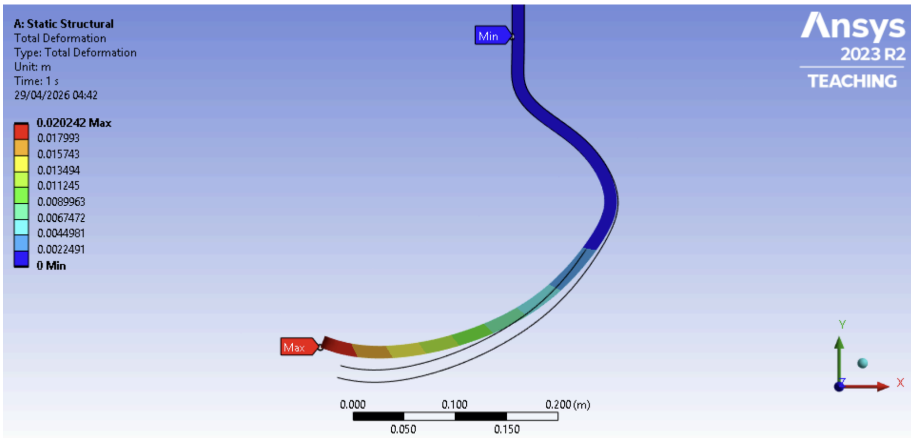

Figure 17: Deformation Plot with Undeformed Wireframe

The maximum stress was 139.64 MPa, on the convex part of the curve, where it is in tension. Stress distribution follows the same pattern as before.

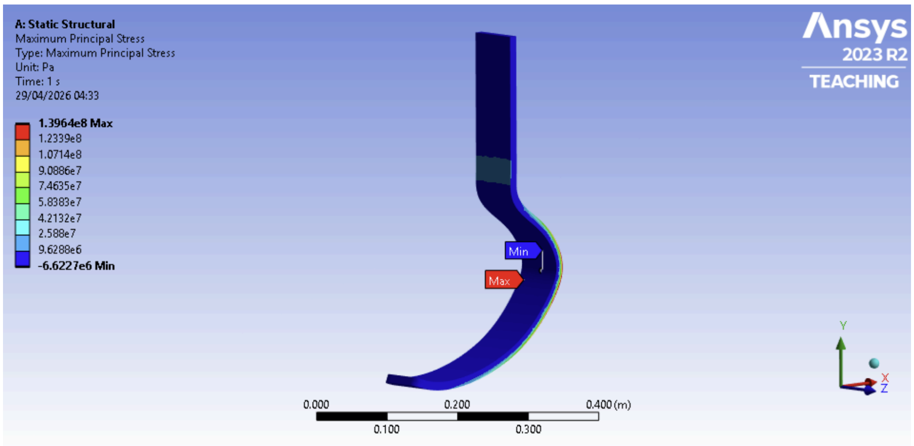

Figure 18: Maximum Principle Stress Plot - Front

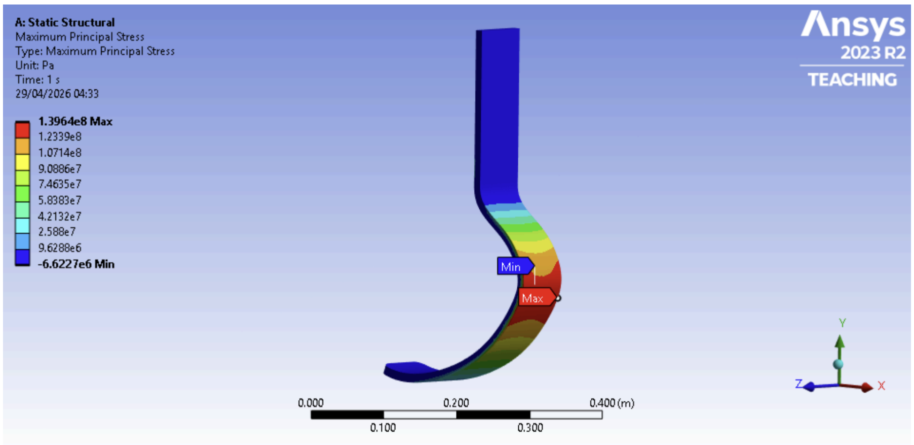

Figure 19: Maximum Principle Stress Plot - Back

Applying the safety factor of 3:

$$139.64 \times 3 = 418.92 \text{ MPa}$$

The effective design stress of 418.92 MPa is below 600 MPa, with a margin of approximately 30%. The redesign passes the set fatigue criteria.

3.2.4 Fundamental Frequency

The fundamental frequency (Mode 1) was 55.252 Hz, passing the 50 Hz requirement. It is roughly twice the previous value of 26.661 Hz, which aligns with the 2× increase predicted by the hand calculation. The Mode 1 shape is the same as before.

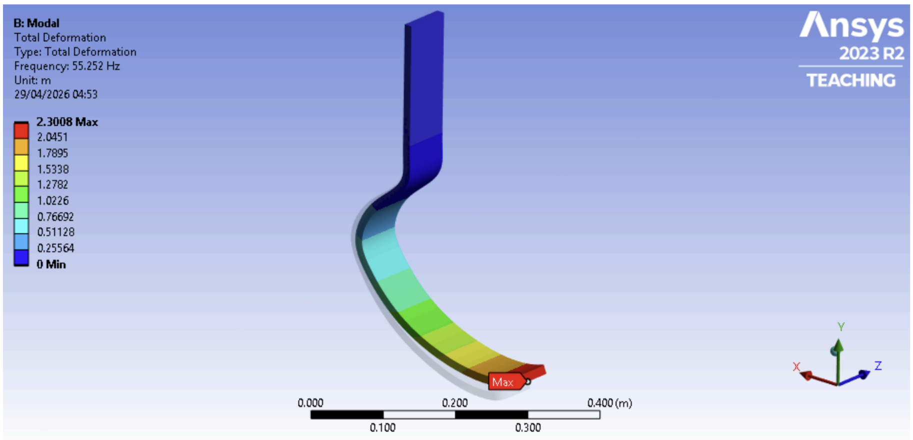

Figure 20: Modal Analysis - Mode 1

The iterated design passes all three criteria: mass under 1kg at 964.16g, fatigue with an effective design stress of 418.92 MPa under the 600 MPa UTS, and a fundamental frequency of 55.252 Hz above the 50 Hz threshold.

3.3 Iterated Design - Aluminium Alloy

The iterated design geometry was kept the same, and the material was switched to aluminium alloy. The same 5 mm + 2 mm mesh was used, since the geometry and loading are identical to those of the CFRP iterated design. The mass increased to 1.669 kg, well above the 1 kg criterion.

3.3.1 Deformation and Static Stress

The maximum deformation was 19.64 mm, at the toe, close to the CFRP value of 20.241 mm. This is expected, as aluminium's Young's modulus (71 GPa) is very similar to the CFRP value (70 GPa). Maximum von Mises stress was 165.78 MPa. This isn't directly comparable to the CFRP iterated design's 139.64 MPa, since that was the maximum principal stress.

3.3.2 Fatigue

The minimum fatigue life was 5.67×10^7 (56.7 million) cycles, far above the 1,000,000-cycle requirement. Therefore, the blade passes the fatigue criteria.

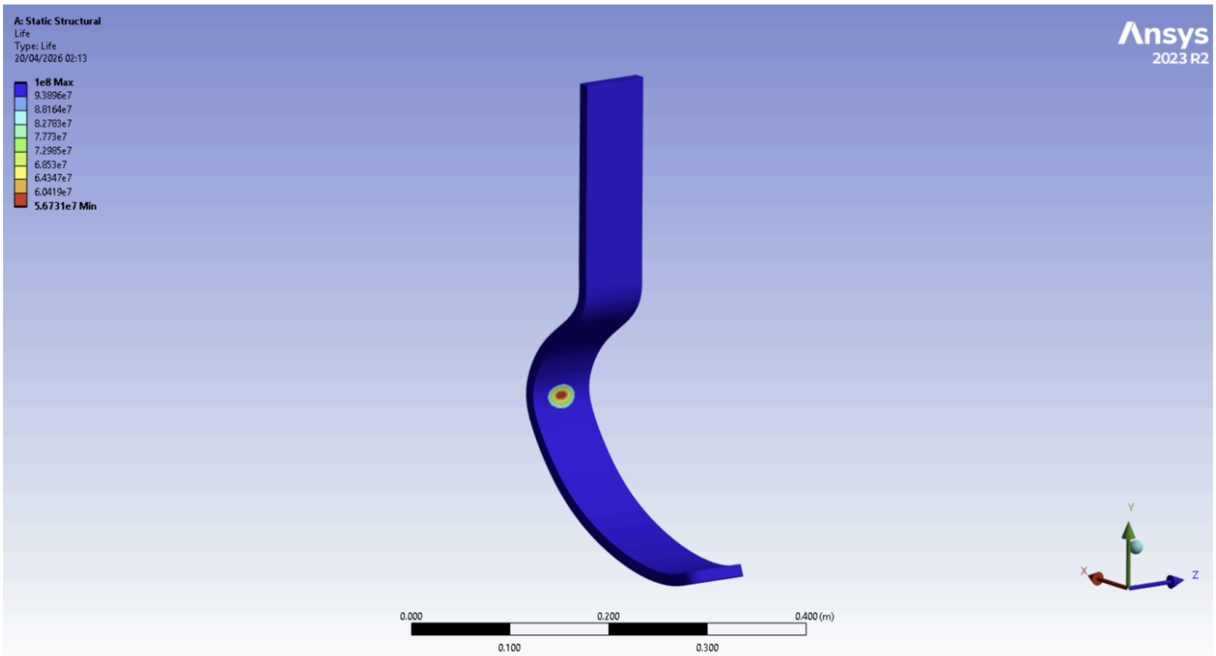

Figure 21: Fatigue Life Analysis

The failure is concentrated at a small spot on the inner concave face; the rest of the blade is at the maximum reportable life of 1×10^8 cycles.

3.3.3 Fundamental Frequency

The fundamental frequency was 42.902 Hz, lower than the CFRP iterated design's 55.252 Hz, and failing the criteria.

The drop is consistent with the cantilever tip-mass formula. Taking the ratio against the CFRP design as a sanity check, changing only E and m:

$$\frac{f_2}{f_1} = \sqrt{\frac{E_2/E_1}{m_2/m_1}} = \sqrt{\frac{1.014}{1.731}} = 0.765$$

This means $f_2 = 0.765 \times 55.252 = 42.27$ Hz, within 1.5% of the result of 42.902 Hz, validating the simulation.

The aluminium iterated design passes fatigue at 5.67×10^7 cycles but fails the lightweight criterion at 1.669 kg and the frequency criterion at 42.902 Hz.

4 Discussion

4.1 Modelling Error

Linear static FEA assumes linear elastic behaviour and small deformations, with the stiffness matrix calculated from the undeformed geometry. This doesn't hold for the initial design, which deflected 75% of its height; the 398 mm result is the output of equations that don't apply at this scale. The redesign deflects 20 mm (4% of height), within the small-deflection range. Enabling Large Deflection in Ansys would have given a more meaningful figure for the initial design, but it already fails on stress and frequency.

CFRP was modelled as isotropic, but in reality, it's orthotropic, with stiffness and strength dependent on fibre direction. Tsai-Wu is a composite failure model that accounts for orientation and would have been the better approach, but it requires more data and the Ansys Composite PrepPost tools, which were out of scope. Maximum principal stress was used as the closest valid alternative for a brittle isotropic material. The pass/fail outcomes are likely correct, but the stress values are not precise.

Friction was applied rearward to model the landing phase. The combination of vertical load and rearward friction produces a larger bending moment about the support than forward friction alone would, since both act in the same rotational sense about the support [6], making this the more conservative case. A transient analysis over the full running phase would be more accurate, since vertical and horizontal forces vary together and their peaks may not coincide. Friction is less than a quarter of the normal force, and the iterated design has a 30% fatigue margin, so this simplification likely doesn't change the outcome.

4.2 Discretisation Error

The convergence studies handled this. Both geometries converged at a global element size of 5 mm with 2 mm local refinement in the highest-stress region. Aspect ratio and Jacobian ratio were tracked at each step; both improved with refinement and were good on the converged mesh. The maximum aspect ratio was the only metric that stayed high in the initial design, due to the thin cross-section forcing stretched elements through the blade. The geometry and loading are symmetric, so a halved model would have reduced the element count, but the full model solved quickly enough that the added setup was not justified.

4.3 Numerical Error

Ansys uses double precision by default, so numerical errors from floating-point rounding are negligible compared to modelling and discretisation errors.

5 Conclusion

The pass/fail outcomes hold up despite the simplifications. The modelling errors discussed would change the size of the margins, but not which side of the line each design sits on.

Design	Mass	Fatigue	Frequency	Result
Initial (CFRP)	518.9 g	2924.1 MPa vs 600 MPa UTS	26.661 Hz	Fail
Iterated (CFRP)	964.16 g	418.92 MPa vs 600 MPa UTS	55.252 Hz	Pass
Iterated (Aluminium)	1.669 kg	5.67×10^7 cycles	42.902 Hz	Fail

Table 12: Final Results Comparison

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[7] Grammarly. *Grammarly.* Grammarly.com, 2025. URL: `https://app.grammarly.com/` (visited on 04/30/2026).

Statement on Use of Writing Tools

Grammarly [7] was used throughout the writing of this report to improve spelling, grammar, punctuation, and readability. It was only used to improve the clarity of my own work. All technical analysis and design decisions are my own.